



DEVELOPMENT AND CHARACTERIZATION OF BIO-COMPOSITES FROM WASTE AQUATIC BIOMASS FOR SUSTAINABLE INSULATION

B.Tech Final Year Project Presentation

Suchetan Karloopia (2022BCHE019) | Miran Haider (2022BCHE021) |

Akshita Sen (2022BCHE037)

Under the guidance of **Dr. Fasil Qayoom Mir**
Department of Chemical Engineering

National Institute of Technology Srinagar

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01

Introduction

Background, problem context and motivation



Background & Problem Context

The Building Sector Challenge

Building sector accounts for substantial global energy use and GHG emissions

Conventional insulation (EPS, PU, glass wool) that are petroleum derived, non biodegradable.

Growing emphasis on **circular economy** driving bio-based alternatives

The Dal Lake Problem

Dal Lake one of the **wetland** (~18 km²) in India, severely eutrophied

Explosive weed growth: *water hyacinth* and *water lily*

De-weeded biomass which is freely available waste stream



Fig. 1: Dal Lake, Srinagar – collection site of raw biomass

Central Focus: Converting invasive aquatic weed into useful engineering product simultaneously addresses two problems which provides economic incentive for weed removal and supplies renewable raw material for sustainable insulation.

Motivation & Problem Statement

01

Environmental Concern

Petroleum-based synthetic insulation materials pose serious environmental and health drawbacks

02

Waste Management

Urgent need to manage and valorize the invasive aquatic weed burden of Dal Lake

03

Demonstrated Potential

Water-Lily fibre has shown low-thermal-conductivity potential in recent literature

Problem Statement

Waste aquatic biomass is a promising sustainable alternative, but direct application is hindered by three deficiencies:

- (1) **Low mechanical strength** = panels fracture under minimal load
- (2) **High moisture absorption** = water displaces insulating air, degrading thermal performance
- (3) **Susceptibility to fire** = lignocellulosic fibres are inherently flammable

Solution Approach: Develop engineered bio-composites with a natural binder (corn starch) and appropriate processing route to ensure structural integrity, dimensional stability and insulation performance while retaining biodegradability.

Project Objectives

01 To Develop Eco-Friendly Bio-Composite Insulation Panels

Fabricate fully bio-based insulation panels using waste aquatic biomass (primarily water hyacinth) from Dal Lake, Srinagar, bound with food-grade corn starch and eliminating all synthetic polymers and chemical crosslinkers.

02 To Optimize Characteristics Through Systematic Testing

Characterize and optimize the physico-mechanical-thermal properties through systematic variation of particle size and biomass-to-binder ratio, targeting insulation-grade thermal conductivity with adequate mechanical strength.

Scope of the Work

- Laboratory-scale fabrication of four bio-composite formulations with varying particle size and starch content
- Physico-mechanical-thermal characterization: MC, bulk density, WA, UCT strength, thermal conductivity
- Microstructural analysis via SEM for S2 and S3 formulations
- Flammability testing, hydrophobic treatments and full LCA , outside present scope

Incorporation of Suggestions

Key improvements incorporated based on mid-term evaluation feedback

Binder System Optimization

Shifted from initial single formulation to systematic variation of biomass:starch ratios (90:10 and 70:30) to identify optimal binder content and quantify the strength-moisture-thermal trade-off.

Enhanced Mechanical Testing

Added unconfined compressive strength testing using Baker Type K12 apparatus to quantify load-bearing capacity (q_u , S_u) beyond qualitative observations of panel integrity.

Particle Size Classification

Introduced ASTM E11 sieve classification into coarse (>3 mm) and fine (1.0–1.5 mm) fractions to systematically study the effect of particle size on packing density, inter-particle bonding and final properties.

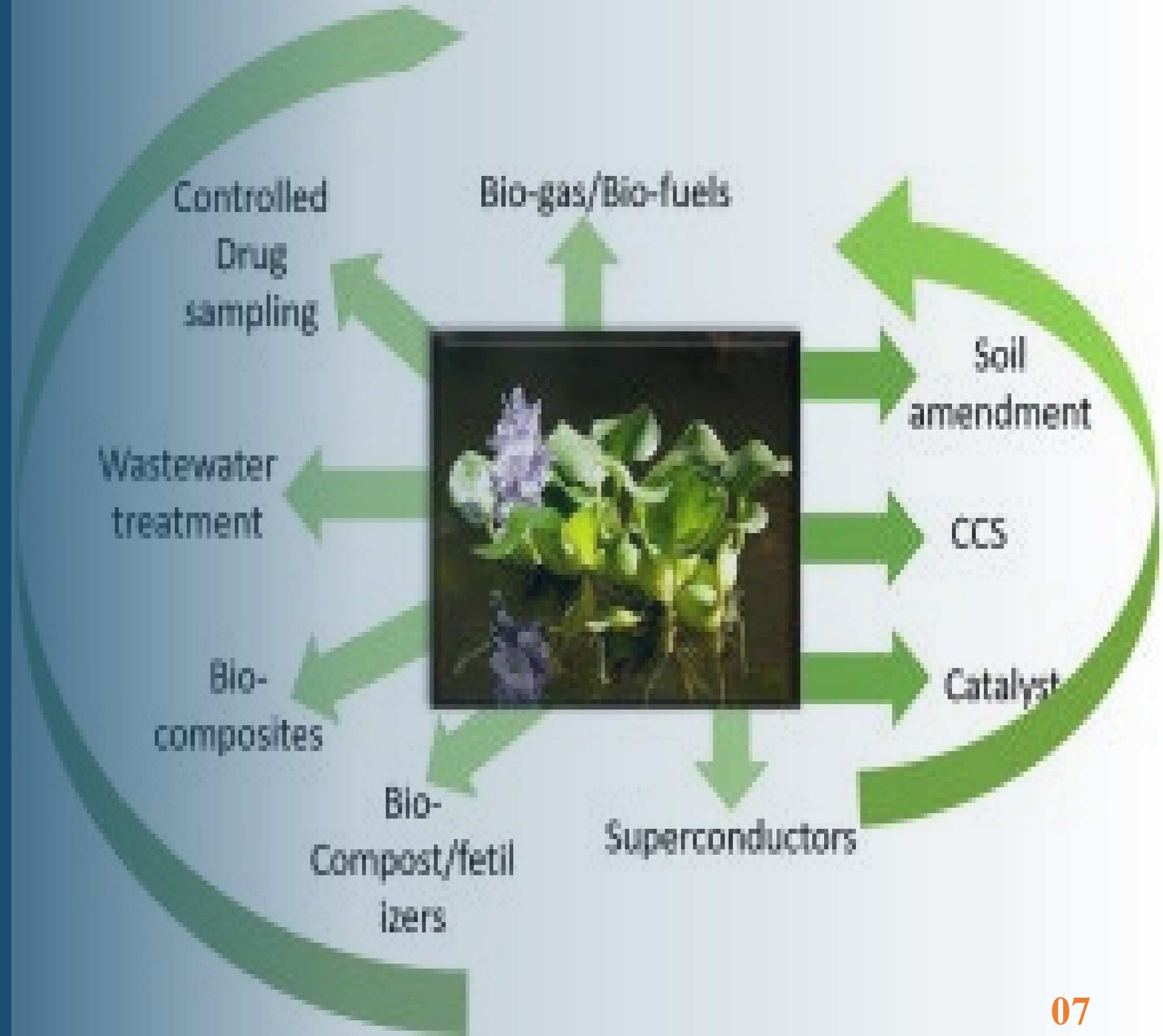
Statistical Rigor & Reproducibility

Standardized all characterization tests to $n=3$ replicates with error bars (± 1 SD) and goodness-of-fit (S_{yx}) reporting for thermal conductivity measurements, ensuring statistically reliable conclusions.

02

Experimental Setup & Methodology

Raw materials, fabrication process, characterization methods



Raw Materials & Biomass Preparation

Materials Used

Material	Category	Function / Rationale
Water lily (<i>Nymphaea spp.</i>)	Primary biomass	Supplementary aquatic biomass reinforcement
Water hyacinth (<i>E. crassipes</i>)	Secondary biomass	Fibrous, cellulose-rich reinforcement and porous insulating phase
Food-grade corn starch	Natural binder	Biodegradable matrix binding fibres after gelatinisation
Water	Processing medium	Disperses biomass and gelatinises starch

Biomass Preparation Steps

1. Collection from Dal Lake + washing to remove debris
2. Sun drying 5–7 days (green → tan/brown)
3. Oven drying at 103±2°C for 24 h
4. Size reduction via mixer-grinder

Particle Size Classification (ASTM E11)

Coarse Fraction Size: > 3 mm (finer than No. 7 mesh) Structure: Open, weakly interlocked	Fine Fraction Size: 1.0–1.5 mm (No. 12–No. 18 mesh) Structure: Larger surface area, more cohesive panels
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Note: No synthetic polymers, resins or chemical crosslinkers were employed at any stage, ensuring the final composite is fully bio-based and biodegradable.

Binder Preparation & Composite Fabrication

Corn Starch Gelatinisation

- Starch–water mixture heated to **80–90°C** with magnetic stirrer
- Stirred until translucent, viscous gel formed
- Gelatinised paste used for S3 (90:10) and S4 (70:30)
- S1, S2: water only (15%) as processing aid

Moulding & Curing

- **Flat panels:** Steel moulds (50 × 50 mm)
- **Cylinders (UCT):** Stainless Steel moulds (D=30 mm, H=25 mm)
- Cold–pressed → demoulded → oven–dried at **103±2°C** (~24 h)

Key Design Logic: Systematically vary particle size (coarse vs. fine) and binder content (0%, 10%, 30% starch) to study the fundamental trade-off between mechanical strength, moisture sensitivity, and thermal insulation performance.

Four Formulations

Sample	Particle Size	Composition	Observation
S1	Coarse >3 mm	Biomass + 15% water	Fractured on demoulding
S2	Fine 1.0–1.5 mm	Biomass + 15% water	Better cohesion; intact panel
S3	Fine 1.0–1.5 mm	90:10 biomass:starch	Improved binding; best balance
S4	Fine 1.0–1.5 mm	70:30 biomass:starch	Dense, compact panel

Process Flowsheet

{ MANDATORY }

S1

Raw Material Collection

Water hyacinth (*Eichhornia crassipes*) and **water lily** (*Nymphaea* spp.) harvested manually from **Dal Lake, Kashmir**. Petioles, stems, and leaves collected, washed with water to remove mud, debris, and biological contaminants.



S2

Sun Drying

Washed biomass spread on clean surfaces and **sun-dried for 5–7 days** under natural sunlight. Moisture removal reduces microbial activity. Observable colour change from green → tan/brown indicates moisture loss.



S3

Oven Drying

Sun-dried biomass transferred to a **drying oven at 105°C for 24 hours** to remove residual moisture and standardize moisture content prior to grinding. (ASTM D4442)



Grinding

S4

Oven-dried biomass ground using a **mixer grinder** (Organic Chemistry Lab, under Prof. Tabassum Ara, Dept. of Chemistry, NIT Srinagar) to reduce particle size for composite fabrication.



Sieving & Particle Size Classification

S5

Ground biomass sieved using a **Sethi Standard Test Sieve shaker** (ASTM E11)

to classify particles:

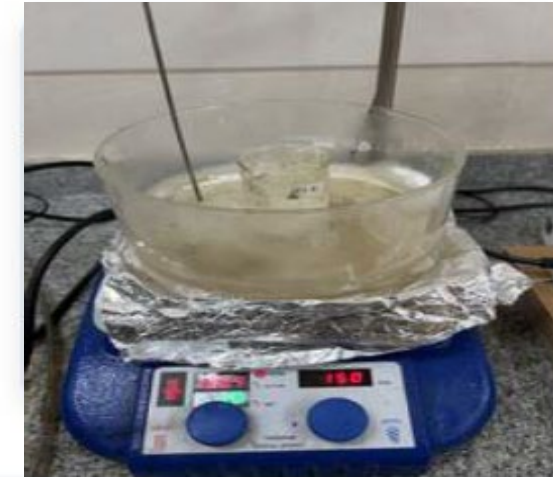
- **Coarse fraction:** >3 mm (< ASTM No. 7 mesh)
- **Fine fraction:** 1.0–1.5 mm (ASTM No. 12–18 mesh)



S6

Binder Preparation (Parallel Step)

Corn starch gelatinized on a **Tarsons Digital Spinot** hot plate with **magnetic stirrer** — temperature: 70°C, stirring: 150 RPM. Starch dissolved in water and heated until gel forms. Used as natural binder for S3 and S4.



S7

Composite Mixing & Mould Filling

Biomass particles mixed with binder/water in defined ratios (see sample variants below) and packed into **wooden moulds** of defined dimensions.

S8

Hydraulic Pressing

Filled moulds compressed using a **Hydraulic Press Model M-30** (Technosearch Instruments) to consolidate composite mixture, reduce porosity, and improve inter-particle bonding.



↓ DEMOULDING → 4 COMPOSITE SAMPLES FABRICATED ↓



SAMPLE 1 (S1)

Coarse particles (>3 mm)
Biomass + 15% water (no binder)
5×5×0.6 cm

⚠️ **Fractured on demoulding**



SAMPLE 2 (S2)

Fine particles (1–1.5 mm)
Biomass + 15% water (no binder)
5×5×0.6 cm

✓ **Intact panel**



SAMPLE 3 (S3)

Fine particles (1–1.5 mm)
Biomass:Starch = 90:10
5×5×0.5 cm

✓ **Improved surface**



SAMPLE 4 (S4)

Fine particles (1–1.5 mm)
Biomass:Starch = 70:30
4.8×4.8×0.3 cm

✓ **Compact & dense**



Sample s2

8:37 PM

Sample 2 (S2)

Fine particles (1–1.5 mm)
Biomass + 15% water (no binder)
Cylinder, D 30 × H 25 mm

✓ **Intact specimen**



Sample s3

8:37 PM

Sample 3 (S3)

Fine particles (1–1.5 mm)
Biomass:Starch = 90:10
Cylinder, D 30 × H 25 mm

✓ **Improved surface**



Sample s4

8:38 PM

Sample 4 (S4)

Fine particles (1–1.5 mm)
Biomass:Starch = 70:30
Cylinder, D 30 × H 25 mm

✓ **Compact & dense**

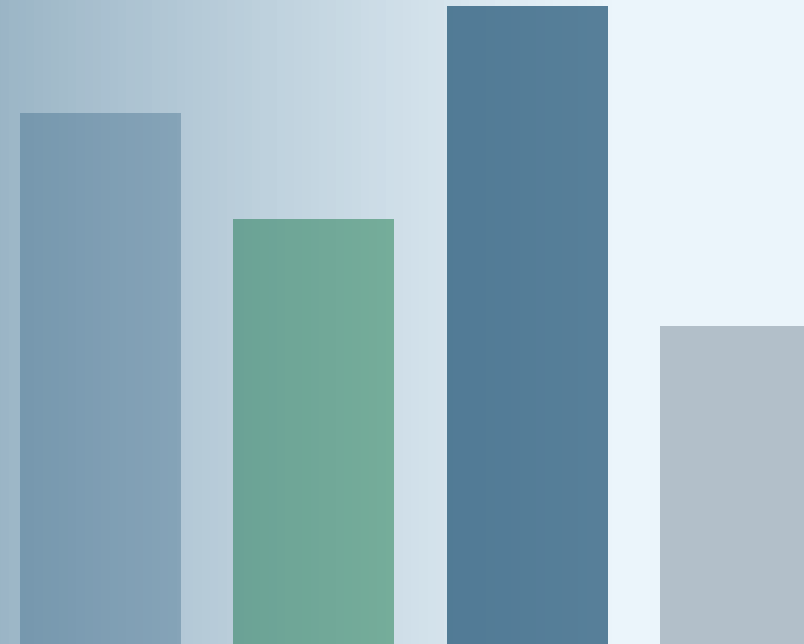
Characterization Methods

Test	Standard / Equipment	Principle / Formula	Replicates
Moisture Content	ASTM D4442, Gravimetric	$MC = (W_i - W_f) / W_i \times 100$	n = 3
Bulk Density	ASTM D1037	$\rho = \text{mass} / (L \times B \times H)$	n = 3
Water Absorption	ASTM D570, 2-hour soak	$WA = (W_{\text{wet}} - W_{\text{dry}}) / W_{\text{dry}} \times 100$	n = 3
UCT Strength	Baker Type K12, C = 2.256 N/div	$q_u = \sigma_{\text{max}}; S_u = q_u/2; \epsilon_f = 16\%$	n = 3
Thermal Conductivity	KD2 Pro TR-3, ASTM D5334 / IEEE 442	Transient line-source method; $S_{yx} < 2.0$	n = 3

03

Results & Discussion

Experimental findings and comparative analysis



Moisture Content & Bulk Density

Fig. 13: Moisture Content (ASTM D4442)

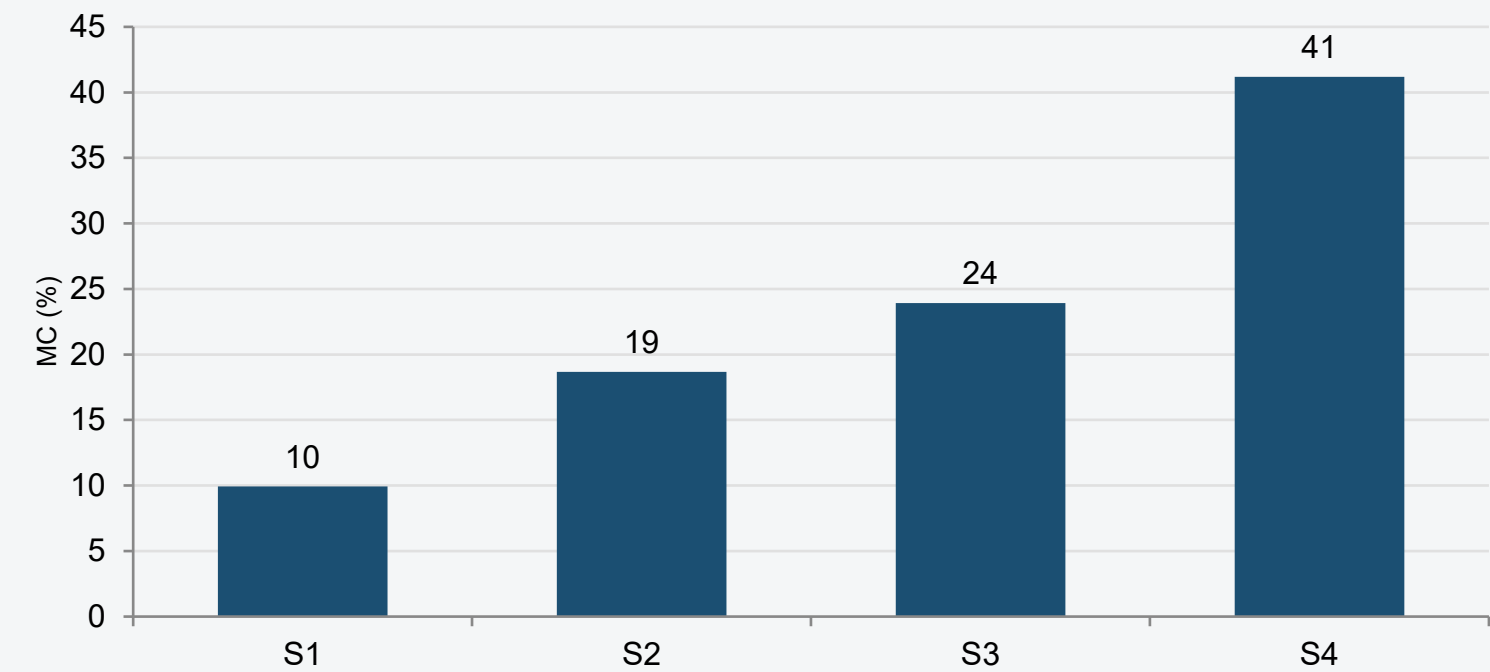
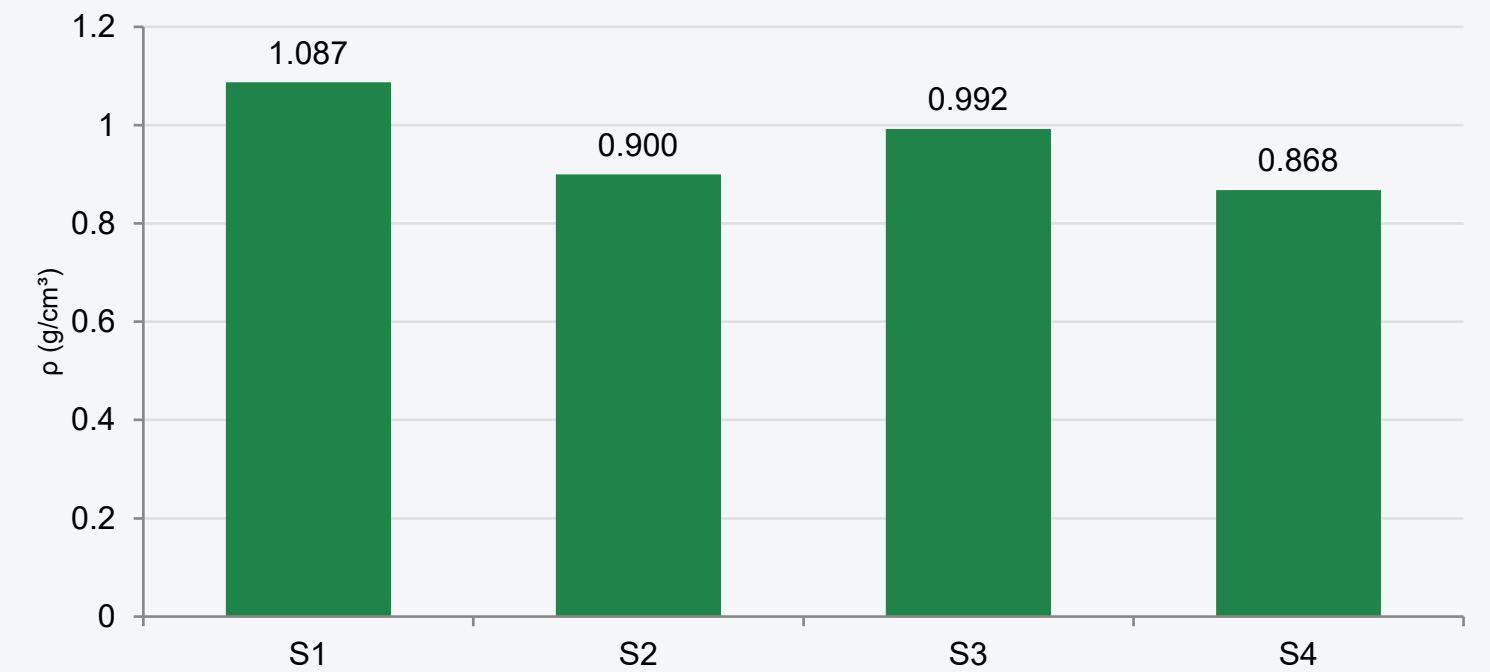


Fig. 14: Bulk Density (ASTM D1037)



Sample	S1	S2	S3	S4
MC (%)	9.94	18.67	23.93	41.18
ρ (g/cm³)	1.087	0.900	0.992	0.868
Key driver	Coarse, porous	Large surface area	Moderate starch	Hygroscopic starch

Key Insight: 10% starch (S3) fills inter-particle voids (ρ = 0.992) without excessive moisture retention (MC = 23.93%) which is a balanced trade-off.

Water Absorption & Compressive Strength

Fig. 15: Water Absorption, 2-hour soak (ASTM D570)

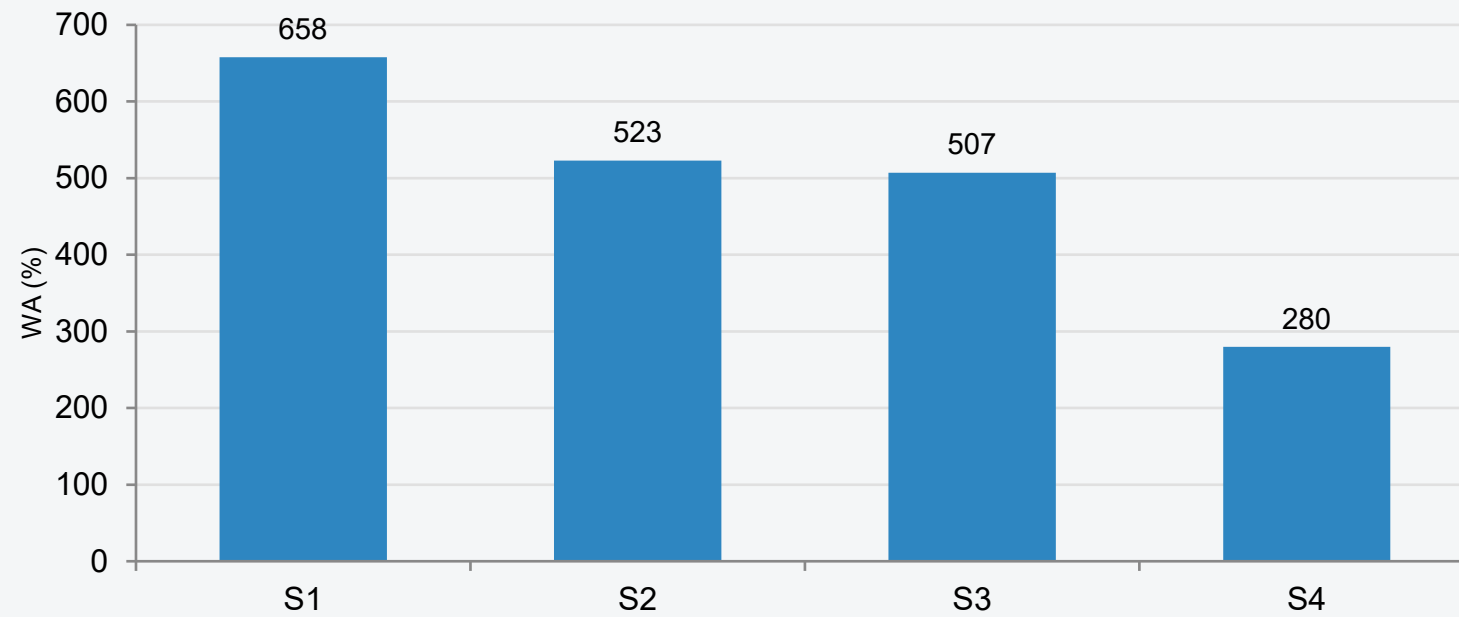
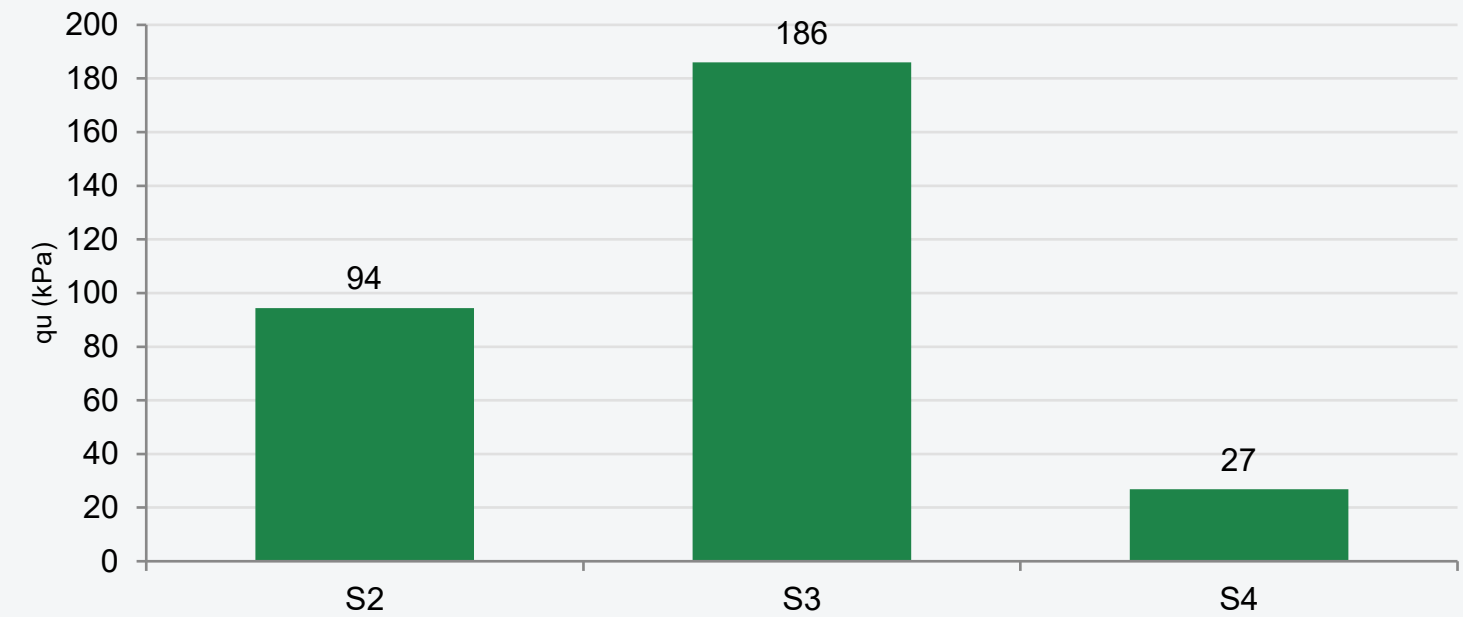


Fig. 17: Unconfined Compressive Strength



Key Findings

Water Absorption: Decreasing trend with increasing starch. S4 (280%) lowest as starch fills pores. All high values need future hydrophobic treatment.

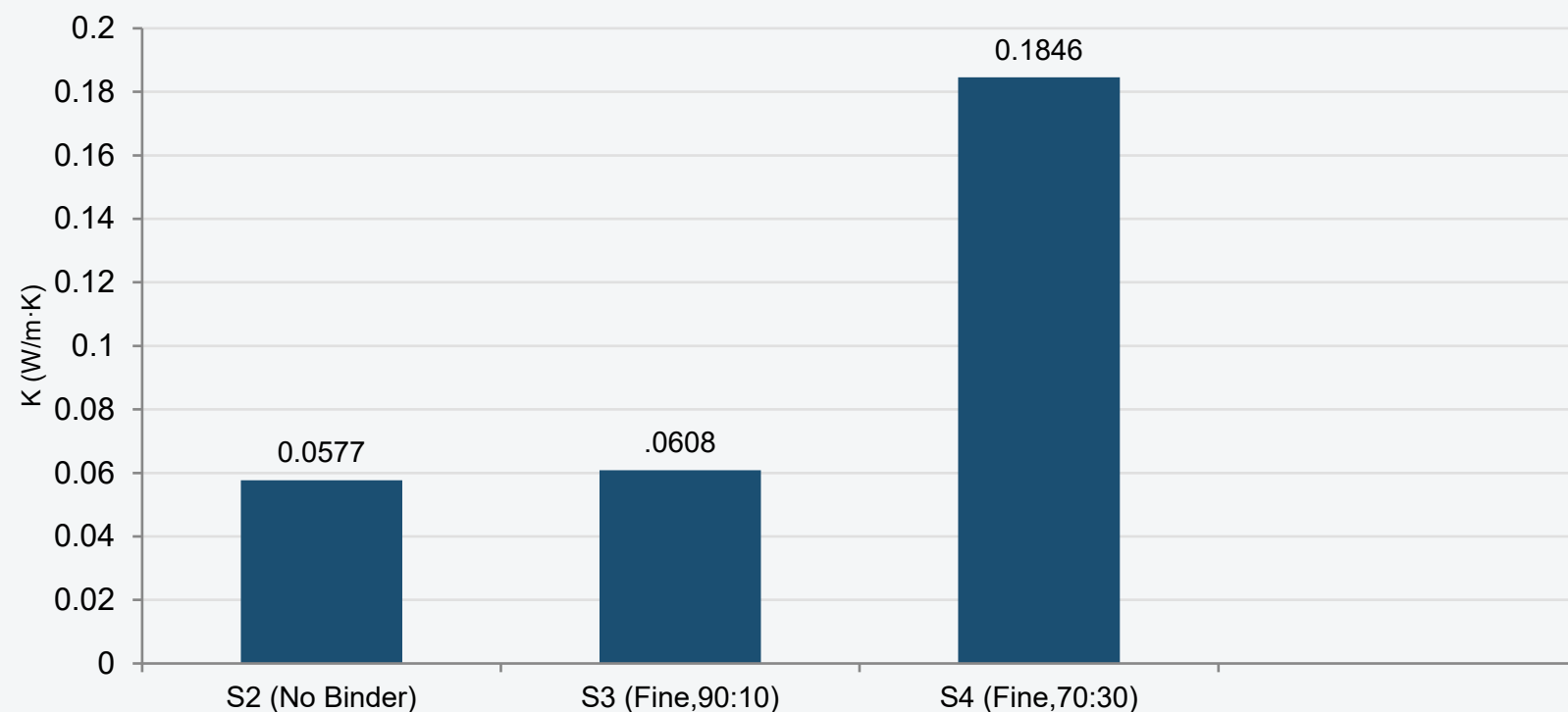
Compressive Strength: S3 highest $q_u = 186.0 \text{ kPa}$ (97% improvement over S2). **10 wt.% corn starch = optimal binder.**

Counter-intuitive: S4 (30% starch) weakest at 26.8 kPa due to excess starch which produces plasticised, moisture-softened matrix.

Uniform failure: All samples at $\epsilon_f = 16\%$ which indicates biomass skeleton governs deformation.

Thermal Conductivity & Optimal Formulation

Fig. 18: Thermal Conductivity vs. ASTM C168 Threshold



Consolidated Results (Mean, n = 3)

Property	S1	S2	S3	S4
MC (%)	9.94	18.67	23.93	41.18
ρ (g/cm³)	1.087	0.900	0.992	0.868
WA (%)	657.9	522.9	507.1	280.0
qu (kPa)	N/A	94.4	186.0	26.8
K (W/m·K)	N/A	0.0577	0.0608	0.1846

Optimal: S3 (90:10 biomass:starch) = Highest strength + insulation; grade K. Green = best.

Why S3 is Optimal

S2 (K = 0.0577 W/m·K): Air-filled voids give excellent insulation but weak mechanically (qu = 94.4 kPa).

S3 (K = 0.0608 W/m·K): Starch menisci bridge fibre junctions → 97% strength improvement. Pores retained → insulation-grade. Best balance.

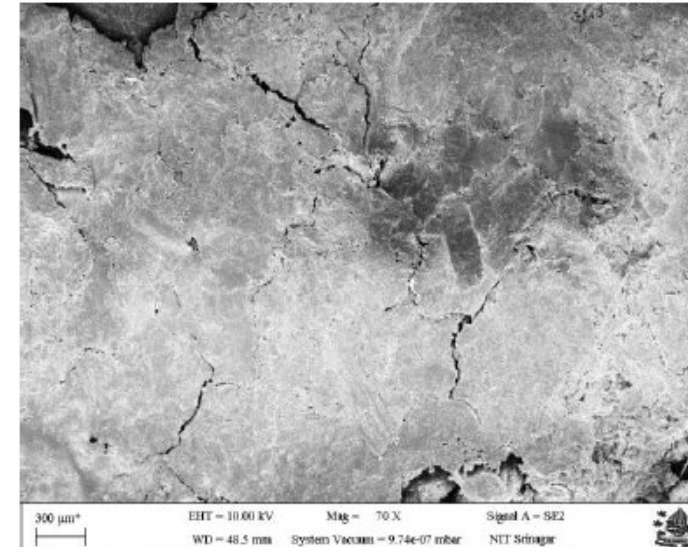
S4 (K = 0.1846 W/m·K): Dense starch fills voids + high moisture displaces air with conductive water → 3× higher K.

Characterization Methods

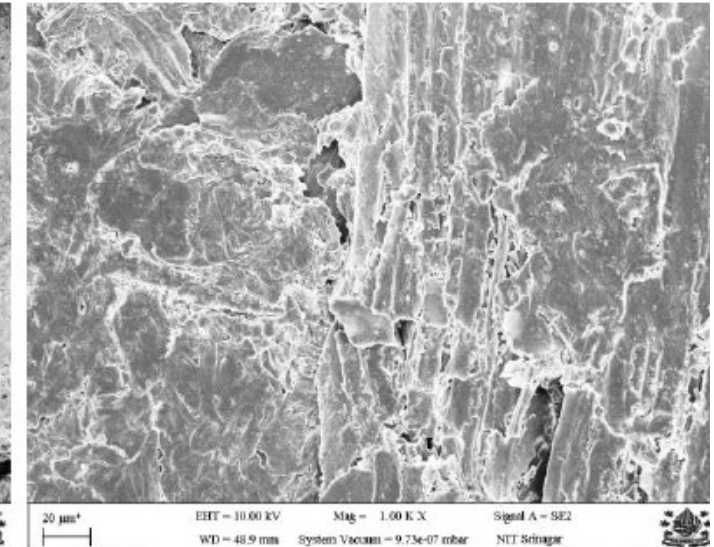
Additional Analysis

SEM Analysis

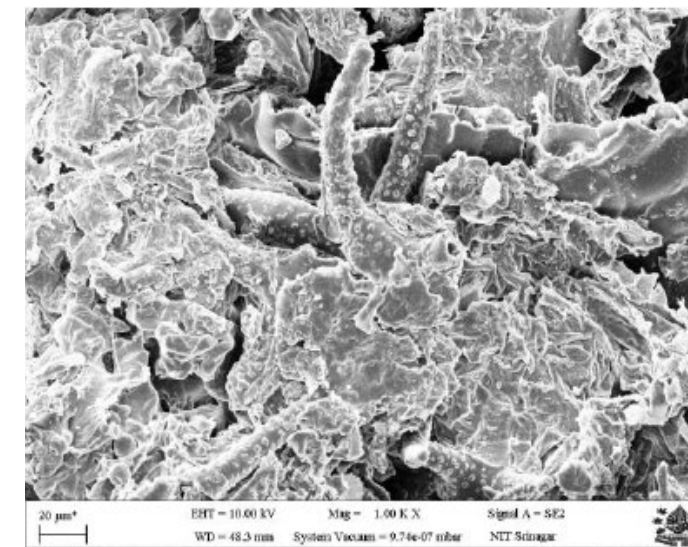
Surface morphology of S1 and S3. Gold sputter-coated, multiple magnifications.



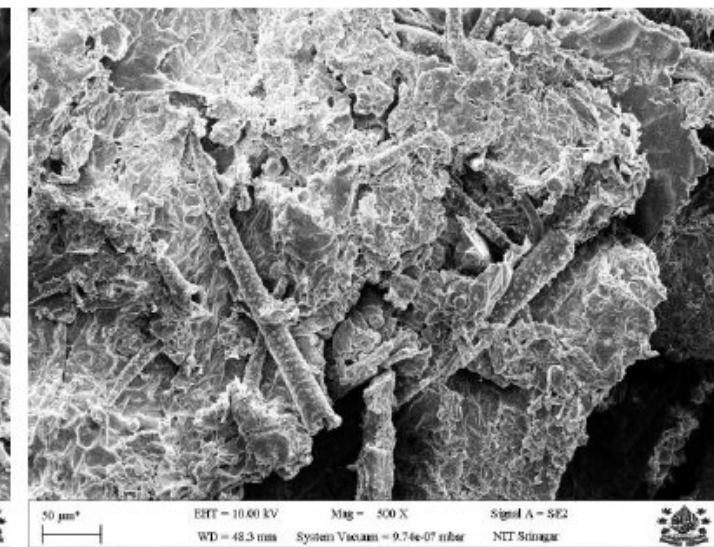
(a) S2 - overview (low magnification)



(b) S2 - fibre network detail



(c) S3 - starch binder bridges



(d) S3 - fibre-matrix interface

SEM Microstructural Analysis

S2 — Fine Binder less Panel

- Open porous fibre network with large inter-particle voids
- Randomly oriented fibres, no preferred alignment
- Uncoated fibre surfaces with exposed cell-wall microfibrils
- Natural tubular cellular cross-sections → trapped-air lumen volume
- **Purely mechanical interlocking** :- no binder bridging between fibres
- Explains low strength and high-water absorption

S3 — Fine 90:10 Starch Panel

- Starch-infiltrated porous network :- binder fills voids selectively
- Gelatinised starch film **coating fibre surfaces**
- Starch menisci bridging inter-fibre junctions as **load-transfer points**
- Good fibre-starch adhesion causing **no pull-out cavities**
- Macro-pores retained despite 10 wt.% starch → insulation-grade K
- Explains 97% strength improvement + maintained insulation

Microstructural Comparison: S1 vs. S3

Feature	S2 (Fine Binder less)	S3 (Fine 90:10 Starch)
Fibre coating	Uncoated, exposed microfibrils	Gelatinised starch film coating
Inter-fibre bonding	Purely mechanical interlocking	Starch menisci bridging
Pore structure	Large inter-particle voids	Macro-pores retained
Load transfer	Fibre-fibre friction only	Starch junctions as load-transfer points
Correlation	Weak, high WA, insulation-grade K	Strong, moderate WA, insulation-grade K

04

Conclusions & Future Work

Key findings and recommendations



Conclusions

- 1 Four bio-composite formulations were successfully fabricated from Dal Lake waste aquatic biomass using **food-grade corn starch binder** which is fully bio-based with **no synthetic polymers or chemical crosslinkers**.
- 2 S2 and S3 achieved **insulation-grade thermal conductivities** of **0.0577 W/m·K** and **0.0608 W/m·K** respectively, both below the **0.065 W/m·K** threshold of ASTM C168.
- 3 **10 wt.% corn starch (S3) identified as optimal binder content** :— delivers **97% strength improvement** ($q_u = 186.0$ kPa) over binder-less panel while maintaining insulation-grade thermal performance ($K = 0.0608$ W/m·K).
- 4 Excess starch (**30%, S4**) degrades both thermal and mechanical properties due to moisture-plasticised matrix ($MC = 41.18\%$, $K = 0.1846$ W/m·K, $q_u = 26.8$ kPa).
- 5 **SEM analysis** confirms starch menisci at fibre junctions act as load-transfer points while preserving insulating pore structure and providing microstructural validation for the macroscopic property trends.

Overall Significance: This work demonstrates the technical feasibility of converting an invasive aquatic weed from Dal Lake into a viable, biodegradable building-insulation material simultaneously addressing waste management and sustainable construction challenges.

Future Work & Scale-Up Strategy

Immediate Future Work

-  **Hydrophobic surface treatment** = beeswax or alkoxysilane coating to reduce water absorption following Chaireh et al. [5]
-  **Flammability testing** = LOI, UL-94 rating, and cone calorimetry to assess fire safety for building applications
-  **Long-term durability assessment** = cyclic humidity and temperature exposure to evaluate in-service performance degradation
-  **Life-cycle assessment (LCA)** = comprehensive environmental impact analysis from raw material extraction to end-of-life

Scale-Up Strategy

Phase 1 – Equipment Transition

From hand-pressing to semi-automatic platen press for consistent compaction

Phase 2 – Format Scaling

Larger moulds: 300×300 mm and 600×600 mm panels

Phase 3 – Pilot Production

Pilot with local weed-management cooperatives

Environmental & Economic Significance

- **Dual-problem solving:** Addresses Dal Lake eutrophication and sustainable building demand simultaneously
- **Zero-waste valorisation:** Converts invasive species into value-added product with no synthetic additives
- **Low embodied energy:** Sun-drying and cold-pressing require minimal energy vs. synthetic insulation
- **End-of-life advantage:** Fully biodegradable which composts naturally unlike petroleum-based alternatives

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(IEEE Format)

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